

Combinatorics

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MATH 31003-01 - Fall 2026

Course: MATH 31003-01

Semester: Fall 2026

Instructor: Gennady Uraltsev

Meetings: Monday, Wednesday, Friday 8:35-9:25 a.m., GEAR 108

Welcome to Combinatorics! We will explore the mathematics of discrete objects: the many ways things can be selected, arranged, connected, and counted. The course covers basic combinatorial techniques, the principle of inclusion-exclusion, generating functions, and recursive relations. Additional topics may include modular arithmetic and applications to encryption, coding theory, graph theory, counting with symmetries, and algebraic structures.

0.1 Start here

- [Syllabus](#): how the course works, grading, materials, and what to expect.
- [Course calendar](#): class dates, current topics, holidays, and exams.
- [Assignments](#): written homework and online questions.
- [Resources](#): references and student-support links.
- [Lecture notes](#): notes organized by course unit and lecture.

Please check the [course calendar](#) for holidays, University deadlines, and topic updates. Course announcements, grades, and submission links are maintained in [Blackboard](#).